University of Greater Manchester

COM4401 · APPLIED COMPUTER SCIENCE FUNDAMENTALS

Cyber-Security Awareness Report

Mini Project 4 · Team Report

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Project team

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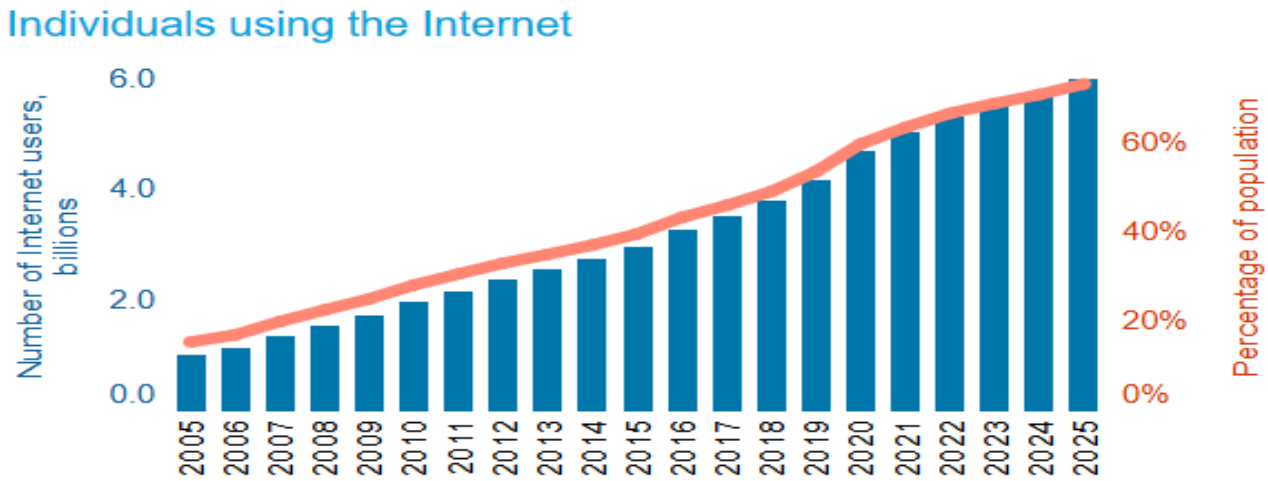
1. Introduction 3
   1.1 Purpose and Importance 4
   1.2 Scope and Intended Audience 4
2. Password Hygiene 4
   2.1 Understanding Password Hygiene 5

2.2 Password Managers and Password Reuse 5
3. Two-Factor and Multi-Factor Authentication 6
3.1 What is 2FA/MFA? 7
3.2 Comparing Authentication Methods 7
3.3 Evidence and Recommended Practice 8
4. Phishing and Social Engineering 9
4.1 Understanding Phishing 10
4.2 Why Phishing Works 10
4.4 Responding to Phishing 11
8. References 12
Appendix A: Team Contribution Record 13
Appendix B: Individual Source Log 14
Appendix C: Peer Review Record 16
Appendix D: Final Submission Checklist 17

1. Introduction

1.1 Purpose and Importance

Cyber-security is the usage of online security methods to protect systems, data, accounts etc. from digital attacks like unauthorised access, malware, brute-forcing, phishing, and other crafty strategies that malicious users may use. As technology usage grows, the number of targets increase for hackers to exploit, and these hackers will also develop stronger attack strategies which highlights the importance of cybersecurity - people and organisations need ways to protect their data in an ever-evolving digital world.



(International Telecommunication Union, 2026).

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2. Password Hygiene

2.1 Understanding Password Hygiene

What does password hygiene mean, and what makes password strong?
Password hygiene is a practical step to prevent accounts from unauthorized access by creating strong passwords, storing, and managing.
Jason Edwards (2025) Strong password is made from a mix of lower- and upper-case letters, special characters numbers, avoid using name of date of birth and it's recommended to be more than 12 characters.

Why has expert guidance shifted from complex character rules towards length and passphrases?
Password unpredictability is dependent on the size of its characters and length *(Kelly T, 2026)*. The computer will take longer to guess if a password is combined from many characters. The recommendation of using upper- and lower-case, special characters and numbers is to set the characters as larger as possible.
For example, if you have 6 characters password in English letters, the computer will need to guess 26 times for each character which will become 308,915,776 times.

2.2 Password Managers and Password Reuse

Password manager is a tool that stores passwords securely by encrypting the password to prevent unauthorized access *(Jason Edwards, 2025)*. Strong Master password is used to access the passwords manager. Password creates and saves passwords for users to use. The user can use a master password to access the tool and help create complex passwords and reduce password reuse *(Grant Collins, 2026).*

Advantage of Password Manager
The advantage of online password managers allows users to access their password remotely. Password manager software can be installed in more than one device and access using master password *(Neil, 2026).*

Password best practice
Best practice to keep your password secure is to change the password after a period, apply password age policy that specifies long user can keep the same password until they must change, avoid reusing the same password, use lower-case, upper-case letter, special characters and numbers with minimum 15 characters to make the password more complex and harder to guess *(Neil, 2026).*

Password monitoring and account security Account Security
Business and organizations check their employee login credential using monitoring tools to see whether similar details have been found in data breach database. This check allows the organization to take action that saves their data from been accessed by cyber criminals, if employee Loging details have been detected in breach database, the organization will review the effected account login activity, change password and apply additional security measure such as passkey, multifactor authentication *(Neil, 2026)*.

Password Attack
Cyber criminals use a technique known as dictionary attack, which involve testing a predefined list of passwords that may include common words, names, predictable patterns, and credentials

obtained from leak databases. Another technique is brute force attack, in which attacker systematically attempts every possible combination of characters until the correct password is identified. Advance in computing technology have increased the effectiveness of brute force attacks, reducing the time required to crack password, especially when password is short or consist of less complex character combinations *(GRC Gate April 2026).*

3. Two-Factor and Multi-Factor Authentication

Section lead: Zidan **Peer reviewed by:** [Name]

3.1 What is 2FA/MFA?

Authentication, the process of proving one's identity, is a highly effective method of security implemented in most of today's technology to prevent the issue of unauthorised access.

There are levels of authentication which are commonly known as 'factors'. They are layers of security that require you to prove your identity at each layer. The more factors an authentication method has, the stronger the security (*Ometov, A. et al. 2018).*

There are three main categories of authentication methods that can be used, which are often combined to create 2FA/MFA authentication (*Zulkarnain, S.I.S. et al. 2013)*:

'Something you know' - passwords, PINs, security questions

'Something you have' - phone/authenticator apps, hardware and security keys, SMS code

'Something you are' - your physical body/biology e.g. facial recognition, fingerprints, eye scanners

Authentication (or SFA, single factor authentication) - typically requires a simple login like a username and password, or a PIN code. This is the first, but weakest level of authentication.

Two-Factor Authentication (or 2FA) - this is the most popular and efficient level of authentication today, providing much stronger security compared to SFA with minimal extra effort to prove yourself. It essentially adds another layer on top of SFA, which can be anything from SMS codes to biometrics. A popular combination nowadays is username/password + SMS code or authenticator app

Multi-Factor Authentication (or MFA) - any authentication method with 2 or more factors. The more factors it has, the stronger security; the trade-off is that it will take longer to validate your identity. MFA is mainly used when the user is accessing highly confidential data/systems like medical systems, corporate finances, etc. A possible combination can be username/password + security key + facial recognition.

3.2 Comparing Authentication Methods

Method	How it works	Strengths	Limitations
SMS code	Sends a code to your mobile number to confirm your identity. (Jain, 2024)	Simple and easy to use, provides efficient security, requires minimal user effort (unless you haven't provided a phone number to the servicer), suitable for most users. Minimal cost compared to other authentication methods like keys, no	Cyber criminals may send a false message to known phone numbers, designing the message to look like it has come from a trusted source. The user may end up revealing sensitive information if they are not careful.

Method	How it works	Strengths	Limitations
		external hardware needed.	
Authenticator app	The app will generate a code, which the servicer requires the user to enter after they have entered the basic credentials to confirm user identity. (Trevino, 2023)	Secure, no hacker interception possible as codes are generated on the user's local device, e.g. phone. Some apps will even reset the code every 30-60 seconds for maximum security, making unauthorized access highly unlikely.	Losing your local device will make it very difficult to get access back. Unlike SMS, where codes are sent to your actual phone number rather than the physical device.
Hardware / security key	A physical device that allows access to your account securely. Can also be used when accessing highly confidential systems. Devices can include NFC and USB. (D'Andrea, 2023)	One of the most reliable and secure methods of authentication. You physically hold a small device that contains your security key, so the chances of unauthorized access are very slim. There is no way for someone to 'hack' into the account or gain access, because the only way is via the physical device.	Despite slim chances of unauthorized access using this method, it is not 100% secure. In the end, they are small physical objects, making them susceptible to theft/loss. Other than this, it essentially has no security drawbacks. Its biggest limitation does not lie on its security, but its compatibility. Global companies like Google and Microsoft support them, but the same cannot be said for smaller/local ones. It can also be quite expensive, requiring between £20 and £75 for each user.

Table 2: Comparison of authentication methods (Source: [citation])

3.3 Evidence and Recommended Practice

There is various evidence that exist to prove the effectiveness of multi-factor authentication - for example the implementation of MFA has reduced the odds of unauthorized access by 98% (Hossain and Raza, 2023), even going up to 99.9% nowadays according to Microsoft reports by (Maynes, 2019). This concludes that MFA is a highly reliable method against the dangers of the digital world like malware, brute-force, keylogging, etc.

Though to reap these benefits, MFA must be configured effectively and use the most appropriate methods depending on what is being accessed. Some methods are weaker than others, whereas some are stronger but more strenuous: Security keys, for example, is widely considered to be one of the strongest MFA methods. This is because the user, and only the user, will have possession of a small object (like a USB stick) to give them access to a system, making it extremely difficult for hackers to intercept, hack into, or phish. Its biggest strength however is also its biggest weakness, meaning if the physical object is lost or stolen, it will be very challenging to recover. This is especially dangerous if a malicious user could exploit it somehow, like if they already know the user's login credentials and stole the key on purpose to pass as an authorised login. There is a very small chance of this happening anyway, so I would still say the benefits outweigh the drawbacks. This method is often used when accessing highly confidential systems - for example business financial data, medical records, and government/military information.

4. Phishing and Social Engineering

Section lead: Saif **Peer reviewed by:** [Name]

4.1 Understanding Phishing

You will impersonate someone else and deceive a victim into doing something that hurts them – like giving you their password, opening an infected attachment or making a bad payment. What is important to note here is not how this was done with technology but the social engineering deception.

Classic phishing attacks usually come in via email. They use a credible reason to send a link to a fake login page. It's a numbers game. The attacker sends the same bait to many thousands of people and counts on a few percent to click.

Spar phishing does everything the same as Classic phishing, except rather than sending bait out to thousands of people the attackers do research on specific persons/groups and select them. Typically, they get info from LinkedIn company sites, etc., where previously breached data may have been stored. The bait sent might be based on a real project, colleague name or conference the target attended lately. There really isn't any difference between spear phishing and Classic phishing just amount of effort /degree. Spear phishing has much greater chance of success since the bait being sent is reliable to the individual receiving it and it's harder for spam filters to find because there won't be enough mail pattern for signature-based defences to look for.

Smish and vishing can best be described as delivering phishing through a different medium then phishing. Smishing is phishing over SMS/Text. It takes advantage of fact that people view texts as more personal/urgent than emails. Messages about package delivery failure or unpaid toll are generally too short to study well, and links are often shortened so final destination is invisible. Additionally mobile devices provide less space to review sender's id/address before clicking. Vishing adds live human element to phishing. Call claiming that person is Fraud department at bank creates pressure, allows attacker to answer objections in real time and steer conversation. Caller id can be easily manipulated to appear as if coming from phone number of valid contact. Because verbal conversation seems more authentic than written message, victims of these scams may resist urge to pause and check stuff like they would when responding to written message

4.2 Why Phishing Works

Phishing works by using humans and not technology. Criminals usually represent a company or person they know you trust and may even be familiar to you from seeing them on social media, other forms of public communications. Since there are millions of pieces of personally identifiable information made available through social media, the criminal can leverage these for their benefit. Using Cialdini’s influence principles (Authority, Social Proof etc.), this makes your decision to give out sensitive information faster and less thoughtful. Most often in phishing emails, the criminals use at least one of Cialdini’s influence principles combined with a sense of urgency and/or time constraint/fear (Your account will be closed today), making it impossible for you to verify whether the email was legitimate or not.

4.3 Common Warning Signs

There are many obvious indicators in past scams that can be easily identified. Today's scams are often presented as being much more polished and professional than those previously seen. Proper spelling, grammar, formatting, etc. have now become options available to many scams. When identifying a suspicious communication through a message or telephone call, I would recommend using several criteria. No longer should you depend upon just one criterion. Disconnect immediately from the individual if you believe they may be fraudulent. Assess the language used by them while communicating with this individual. .

Warning sign	Why it may be suspicious	What the user should do
Urgent demand to act now	Pushes people and rushes them so they can't check the messages carefully	Stop what you are doing contact the organisation using a phone number you already know or website
Unexpected link or QR code	QR codes are starting to be used in scam websites with the destination being hard to see	Do not click or scan. Report the message instead (NCSC, no date b)
Odd or unfamiliar sender, or details that look wrong	Scammers can copy trusted brands so designs that are convincing shouldn't be taken as proof	Check the actual sender address, not just the display name
Request for passwords, bank details or personal information	Genuine organisations rarely ask for these through unsolicited messages	Do not reply. Go directly to the official website if you need to check

4.4 Responding to Phishing

if you receive an email that appears to be a scam, do not reply to the email and/or click on any links. the NCSC accepts reports from members of the public about emails, texts, websites and telephone calls which appear to be fraudulent. NCSC will investigate the claims and remove the offending website. (NCSC, no date a; NCSC, no date c) If you have already suffered financially or otherwise due to responding to such an email or if your account details have been accessed as a result of responding to such an email, then you are advised to also make a formal complaint to the relevant authorities. (NCSC, no date a) In addition, staff working within organisations should promptly notify their organisation's IT department of suspected phishing attempts in order for their fellow colleagues who may have also received similar emails to be notified.

Cialdini, R.B. (2007) *Influence: The Psychology of Persuasion*. Rev. edn. New York: Harper Business.

National Cyber Security Centre (no date a) *Phishing: Spot and report scam emails, texts, websites and calls*. Available at: https://www.ncsc.gov.uk/collection/phishing-scams (Accessed: 22 October 2026).

National Cyber Security Centre (no date b) *Spot scams*. Available at: https://www.ncsc.gov.uk/collection/phishing-scams/spot-scams (Accessed: 22 October 2026).

National Cyber Security Centre (no date c) *Report a scam website*. Available at: https://www.ncsc.gov.uk/collection/phishing-scams/report-scam-website (Accessed: 22 October 2026).

Workman, M. (2008) 'Wisecrackers: a theory-grounded investigation of phishing and pretext social engineering threats to information security', *Journal of the American Society for Information Science and Technology*, 59(4), pp. 662–674. doi: 10.1002/asi.20779.

Appendix A: Team Contribution Record

Team member	Sections written	Approx. words	Contribution to intro / conclusion	Sections peer reviewed
Zidan	1, 3	976	Intro	
Saif	4	980		
Farouk	2	700		

Appendix B: Individual Source Log

Each student: at least **5 credible sources**, including at least **1 peer-reviewed** and **2 authoritative professional or government** sources.

Student	*Source (Harvard reference)*	*Type*	*Used in section*
Farouk	Jason Edwards (2025) *Cyber Defence.* ed. [Online]: Wiley. Available from: https://. [Accessed 14.10.2026].	Book	2
Farouk	*Kissell, Joe. (2019) 'Take Control of Your Passwords, 3rd Edition', p. 160. Available at: https://www.oreilly.com/library/view/take-control-of/9781492066408/text/ch001.xhtml (Accessed: 16 October 2026).*	Book	2
Farouk	*Grant Collins (2026) Complete Introduction to Cybersecurity. ed. [Online]: Available from: Complete Introduction to Cybersecurity [Accessed 15.10.26].*	Video	2
Farouk	*Neil, I. (CompTIA® Security+ SY0-801 Certification Guide.4th edition. [online] available from: https://learning-oreilly-com. [Accessed 19.10.26].*	Book	2
Farouk	1. *RC Gate (2026) Ultimate CISSP Certification Preparation course. Video available at: https://learning-oreilly-com.ezproxy.bolton.ac.uk/(accessed: 20.10.2026).*	Book	2
Zidan	*Jain, R. (2024) 'Understanding SMS authentication: Secure your online accounts'. https://www.fyno.io/blog/what-is-sms-authentication-and-how-secure-is-it-clywqr0ic006vnvor732r8tzs/. Accessed: 14 October 2026*	O	3

Student	Source (Harvard reference)	Type	Used in section
Zidan	*Trevino, A. (2023) 'What are authenticator apps and how do they work?' https://www.keepersecurity.com/blog/2023/07/20/what-are-authenticator-apps-and-how-do-they-work/.* *Accessed: 14 October 2026*	O	3
Zidan	*D'Andrea, A. (2023) 'What is a hardware security key and how does it work?' https://www.keepersecurity.com/blog/2023/05/09/what-is-a-hardware-security-key-and-how-does-it-work/.* *Accessed: 14 October 2026*	O	3
Zidan	*Ometov, A. et al. (2018) 'Multi-Factor Authentication: a survey,' Cryptography, 2(1), p. 1. https://doi.org/10.3390/cryptography2010001. Accessed on 16 October 2026*	P	3
Zidan	*Zulkarnain, S.I.S. et al. (2013) 'A review on authentication methods,' HAL (Le Centre Pour La Communication Scientifique Directe) [Preprint]. https://hal.science/hal-00912435v1. Accessed on 16 October 2026*	P	3
Zidan	*Hossain, M.A. and Raza, Md.A. (2023) 'EXPLORING THE EFFECTIVENESS OF MULTIFACTOR AUTHENTICATION IN PREVENTING UNAUTHORIZED ACCESS TO ONLINE BANKING SYSTEMS,' SSRN Electronic Journal [Preprint]. https://doi.org/10.2139/ssrn.5207142. Accessed on 20 October 2026*	P	3
Zidan	*Maynes, M. (2019) One simple action you can take to prevent 99.9 percent of attacks on your accounts. https://www.microsoft.com/en-us/security/blog/2019/08/20/one-simple-action-you-can-take-to-prevent-99-9-percent-of-account-attacks/. Accessed on 20 October 2026*	A	3
Zidan	*International Telecommunication Union (2026) Global ICT statistics: internet connectivity and adoption trends. Geneva: ITU. https://www.itu.int/en/ITU-D/Statistics/Pages/stat/default.aspx Accessed 20 October 2026*	A	1
Mohammed	*Cialdini, R.B. (2007) Influence: The Psychology of Persuasion. Rev. edn. New York: Harper Business.*	O	2
Mohammed	*National Cyber Security Centre (no date a) Phishing: Spot and report scam emails, texts, websites and calls. Available at: https://www.ncsc.gov.uk/collection/phishing-scams (Accessed: 22 October 2026).*	A	1,2,4

Student	Source (Harvard reference)	Type	Used in section
Mohammed	*National Cyber Security Centre (no date b) Spot scams. Available at: https://www.ncsc.gov.uk/collection/phishing-scams/spot-scams (Accessed: 22 October 2026).*	A	3, Table
Mohammed	*National Cyber Security Centre (no date c) Report a scam website. Available at: https://www.ncsc.gov.uk/collection/phishing-scams/report-scam-website (Accessed: 22 October 2026).*	A	4
Mohammed	*Workman, M. (2008) 'Wisecrackers: a theory-grounded investigation of phishing and pretext social engineering threats to information security', Journal of the American Society for Information Science and Technology, 59(4), pp. 662–674. doi: 10.1002/asi.20779.*	P	2

Type: P = peer-reviewed academic · A = authoritative professional / government · O = other credible source